

Global Player Mobility in Football: A League-to-League Network Analysis

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Abstract

This project analyzes global player mobility in professional football using transfer data from the Transfermarkt website. Player moves are represented as a directed, weighted network on league level, where edges capture transfer flows between source and destination competitions across seasons. Four structural hypotheses were tested, concerning market concentration (H1), the bridging role of mid-tier leagues (H2), community structure relative to a degree-preserving null model (H3), and the domestic orientation of lower-tier transfers (H4). The market is found to be only moderately concentrated and becoming less so over time, mid-level leagues do not act as disproportionate bridges in the original betweenness sense, but a revised analysis on a global-standing axis shows that mid-standing top-flight leagues do broker upward, cross-regional flows. Community structure is strong and geographically coherent, and lower-tier transfers are overwhelmingly domestic. The work follows CRISP-DM and is fully reproducible.

1 Introduction

1.1 Motivation

Professional football is a global labour market, where players move between clubs and leagues across countries. According to FIFA's *Global Transfer Report 2025*, clubs worldwide completed 24,558 international transfers in men's professional football during 2025, with transfer fees reaching USD 13.08 billion, more than 50% above the previous year [FIFA, 2026]. These figures show that this labour market is large, volatile, and structurally embedded in professional football.

Despite this scale, public discourse often relies on qualitative descriptions of individual high-profile moves and on recurring labels such as “selling leagues”, “import leagues”, or “stepping-stone pathways”. These labels are rarely defined quantitatively. From a data-science perspective the transfer market is a natural candidate for network analysis. Each transfer can be modelled as a directed event from a source club to a destination club, embedded in a source and destination league, with an exact timestamp. This structure forms a directed, weighted network from which valuable information can be extracted about the structural roles of leagues, the geography of player flows, and the temporal evolution of market concentration.

1.2 Problem Statement & Research Questions

The goal of this project is to model the global transfer market as a directed, weighted network at the league level, and to use this network and appropriate tools to answer four research questions about its structural properties. The unit of analysis is the league (competition); each node is a single league, each directed edge (u, v) represents player transfers from league u to league v , and edge weights are transfer counts aggregated over a defined time window. The data source is Transfermarkt (<https://www.transfermarkt.com>) and the analysis covers 32 seasons, from 1994/95 to 2025/26. The four research questions and hypotheses are as follows:

- **RQ1 – Mobility structure and concentration.** How is player mobility between leagues structured as a directed and weighted network, and how concentrated are incoming transfer

flows across leagues?

H1: Incoming transfer flows are highly concentrated, with a small set of leagues accounting for a large share of all incoming transfers.

- **RQ2 – Centrality and roles.** Which leagues primarily act as senders or receivers of players, and which leagues function as bridges between otherwise weakly connected parts of the network?

H2: Some mid-level leagues exhibit disproportionately high betweenness centrality and act as bridges between regions and/or tier systems.

- **RQ3 – Communities and mobility regions.** Are there clusters of leagues that form strongly connected mobility ecosystems, and how stable are these clusters across seasons?

H3: The network exhibits non-random community structure, with detected communities showing higher modularity than a degree-preserving randomised baseline.

- **RQ4 – Lower-league pathways.** How important are transfers between lower tiers and top tiers, and how do these pathways differ from top-tier mobility?

H4: Transfers originating from lower tiers are more likely to be domestic and upward (from lower to higher tier) within the same country, while cross-border moves are relatively less common compared to top-tier mobility.

The corresponding evaluation metrics are the Herfindahl–Hirschman Index and Top-5 inflow share for H1; the mid-tier share among the top-25 leagues by betweenness centrality, including its lift relative to the mid-tier base rate, for H2; the empirical p -value of observed modularity against a degree-preserving null model for H3; and the weighted domestic share of outgoing transfers by source-tier group for H4. The full KPI definitions and thresholds are reproduced in Section 4.

1.3 Scope and Contribution

The project is primarily a descriptive and structural analysis and does not attempt to predict transfer fees, forecast individual moves, or model player valuations. Its contribution is to characterise the structure of the global league-to-league network, test the four hypotheses, and do so reproducibly across 32 seasons.

1.4 Outline

This report closely follows the CRISP-DM process model [Wirth and Hipp, 2000]. Section 2 reviews prior work on football transfer networks and situates this project’s methodological choices within that literature. Sections 4 to 9 correspond to the six CRISP-DM phases: business understanding (Section 4), data understanding (Section 5), data preparation (Section 6), modelling (Section 7), evaluation (Section 8), and deployment (Section 9). Section 10 concludes with a verdict-by-hypothesis summary and a brief discussion of limitations. An exploratory analysis of loan dynamics is included as Appendix A.

All stochastic steps in the analysis are seeded with the author’s student ID (seed = 12412672) for full reproducibility, and the analysis pipeline is split across two notebooks executed in sequence under a pinned Python environment.

2 Related Work

Research using Transfermarkt football data can be divided into two broad strands. The first focuses on individual player valuation, typically using regression or machine-learning models to predict fees or future prices, whereas the second focuses on network analysis of transfer flows. This project focuses on the second strand and is positioned as follows.

Earlier work on transfer networks operates mostly at the club level. Liu et al. [2016] cover 24 top-class leagues worldwide between 2011 and 2015 and relate club centrality to sportive success. Li et al. [2019] build a club-level network of roughly 470,000 transfers over 1990–2016, involving more than 23,000 clubs across 206 countries and characterise its degree and strength distributions. Matesanz et al. [2018] cover 21 European top-tier leagues over 1996/97–2015/16 and combine network metrics with self-organising maps. Work such as Wand [2022] adds depth by analysing fee-weighted networks of eight top-tier European leagues from 1992–2020. The study finds that the market is not small-world, possibly because clubs primarily trade with financially similar partners, and explicitly mentions that follow-up work should replace fees with transfer counts as edge weights to also include lower-tier leagues. D’Urso et al. [2025] run community detection on the seven top European leagues between 2009/10–2021/22 under both weightings and report that fee weighting clusters the richest clubs cross-nationally, whereas count-weighted edges “faithfully reconstruct the national leagues”.

This project differs from the above in several ways. First, the unit of analysis is the league rather than the club, which surfaces patterns at the national level. Second, the scope covers all professional leagues recorded on Transfermarkt over 32 seasons (1994/95–2025/26), including all lower-tier and academy leagues. Third, the edges are weighted by transfer counts rather than fees.

Methodologically, community detection uses the Leiden algorithm, which improves on the Louvain algorithm by guaranteeing well-connected communities [Traag et al., 2019]. Significance is tested against a degree-preserving null model implemented via random edge rewiring. Temporal trends are tested with the non-parametric Mann–Kendall test [Mann, 1945, Kendall, 1975].

The combination of a league-level, global, count-weighted, multi-decade scope with Leiden-based community detection and a null-model significance test is, to the best of the author’s knowledge, novel.

3 Methodology

3.1 Process model: CRISP-DM

The project closely follows the CRISP-DM process [Wirth and Hipp, 2000]. The six phases map to the remaining sections of this report as follows: business understanding (Section 4), data understanding (Section 5), data preparation (Section 6), modelling (Section 7), evaluation (Section 8), and deployment (Section 9). The proposal’s research questions and hypotheses (H1–H4) and the corresponding KPIs (Table 1) were fixed at the business-understanding phase and carried through unchanged.

3.2 Methods stack

Network structure is described with standard measures on a directed, weighted graph, including weighted in/out strength and distance-weighted betweenness centrality. Community structure is detected with the Leiden algorithm [Traag et al., 2019]. Statistical inference uses the Herfindahl–Hirschman Index (HHI) for concentration (bounded between 0 for perfect equality and 1 for a full monopoly); a degree-preserving null model for the significance of observed modularity; a label-permutation test for the curvature of the H2 bridge-score fit; and the non-parametric Mann–Kendall trend test [Mann, 1945, Kendall, 1975] for temporal trends. The domain-specific lecture *Foundations of International Business* (WU Wien) frames the transfer market as a structured cross-border labour market and motivates the geographic and tier-based hypotheses H3 and H4.

3.3 Experiment design and reproducibility

All stochastic steps are seeded with the author’s student ID, $\text{seed} = 12412672$. The pipeline is split into two notebooks run in sequence: `01_data_preparation.ipynb` produces the cleaned edge lists under `data/prepared/`, and `02_analysis.ipynb` uses them and writes all figures and tables to separate folders. The dataset itself is private.

Robustness is addressed by computing the H1 concentration metrics in four variants (with and without self-loops, with and without loan transfers) and by computing H1 and H3 metrics per season to check temporal stability.

All experiments and code are available at
<https://github.com/timg4/Mobility-in-Global-Football>.

4 Business Understanding

As described in the proposal, the goal of this project is to characterise the structure of the global league-to-league transfer network and test four hypotheses about its concentration, bridging behaviour, community structure, and the domestic orientation of lower-tier transfers. Success is defined ex ante by the KPIs and decision thresholds in Table 1; the results are evaluated against these in Section 8.

Methodological choices (league-level unit of analysis, count-weighted edges, exclusion of pseudo-destinations) and the reproducibility setup (seeded randomness, pinned environment, two-notebook pipeline) are described where they apply, in Sections 6–9.

5 Data Understanding

5.1 Source

The data is directly sourced from Transfermarkt.com and was provided by WU Wien. It contains five tables in total from which four are relevant for this project: the transfers, player, club and market-value table. Among other information, each transfer record links a player, a

H	KPI	Decision rule
H1	HHI of inflows; Top-5 league share; Mann–Kendall trend test per season	“Concentrated” if $HHI > 0.15$ or Top-5 share $> 50\%$; trend significant at $\alpha = 0.05$.
H2	Original: mid-tier share (tier 2–4) among top-25 leagues by betweenness, vs. mid-tier base rate. Revised: inverted-U fit of bridge score against global-standing percentile, with label-permutation null.	Original: “Supported” if lift $\geq 1.5\times$. Revised: curvature $c_2 < 0$ and $p < 0.05$.
H3	Observed modularity Q vs. degree-preserving null model	$p < 0.05$ one-sided.
H4	Weighted domestic share of outgoing transfers by source-tier group	“Supported” if domestic share $> 70\%$ for tier-3 and lower.

Table 1: KPIs and ex-ante decision rules for hypotheses H1–H4.

source and destination club, source and destination competition (league) identifiers, source and destination country identifiers, a transfer type, a fee field, and a date.

5.2 Missing Values and Initial Observations

- **Missing competition IDs.** A large share of transfer records (approximately 300,000) have a missing source and/or destination competition ID, while not a single club ID is missing in the entire dataset.
- **Pseudo-destination encoding.** Inspecting the clubs with the most incoming and outgoing transfers revealed that several club IDs are in fact non-club states with a specific encoding used by Transfermarkt. Club 123 denotes *Retired*, club 515 *Without Club*, club 75 *Unknown Club*, and club 2113 *Career Break*.
- **Missing fees.** The fee field is missing for a large fraction of records (approximately 880,000 transfer records). The fee-label field revealed that most of these are genuine free transfers, but transfer fees in global football are also frequently undisclosed. Missing fees are therefore not missing at random.
- **Encoded dates.** The date field is an integer. Three candidate encodings (days, seconds, milliseconds since 1970-01-01) were tested against a plausible date window. This was done by comparing with three well-known transfers: Neymar to Paris Saint-Germain (2017), Cristiano Ronaldo to Real Madrid (2009), and Ronaldo to Inter (1997). All of them were mapped correctly when using days since 1970-01-01.

5.3 Temporal coverage

The transfer data are sparse and incomplete before the mid-1990s. The dataset also contains implausible outliers in the earliest years and records dated after the handover date, most of which are planned loan returns. Annual coverage indicates that the data become sufficiently reliable from 1995 onwards. The analysis window was therefore fixed from 1 January 1995 to

27 January 2026. Earlier records were dropped because of sparse coverage, and later records were excluded because planned loan returns were uncertain and should not yet be represented as completed transfers.

5.4 Descriptive overview

After cleaning and restriction to the analysis window, the dataset has the figures that can be seen in Table 2.

A total of 921,332 cleaned transfer records fall within the analysis window of which 586,520 have an identified league at both ends and are therefore usable for this analysis. The strict subset spans 672 distinct leagues across 118 countries. Loans account for roughly one in four transfers.

Dimension	Value
Analysis window	1994/95 – 2025/26
Seasons	32
Transfer records (clean, in window)	921,332
Records with both leagues mapped (strict)	586,520
Unique leagues (strict)	672
Unique countries (strict)	118
Loan transfers	249,985 (27.1%)
Permanent transfers	671,347 (72.9%)

Table 2: Descriptive overview of the cleaned dataset.

5.5 Transfer types and fees

The transfer-type field includes, among others, permanent moves, loans, loan returns and free transfers. Loans are a legitimate form of transfer and are therefore included in the analysis. The non-random pattern of missing fees seems to be systematic, and using them in the analysis would be biased against smaller and non-European leagues. Therefore, in the subsequent phases, the network edges are weighted by transfer counts rather than fees, and fees are not used further in the analysis.

6 Data Preparation

The data preparation produces four edge-list files by following the steps below in order.

6.1 Date decoding and window restriction

The integer date field is decoded as days since 1970-01-01 (Section 5). Records before 1 January 1995 are dropped as pre-coverage, and records after the dataset handover (27 January 2026) are dropped as out-of-window. Each remaining transfer is assigned a season using a July cut-off: transfers from July of year Y through June of year $Y+1$ belong to season $Y/Y+1$.

6.2 Pseudo-destination handling

The four club IDs that do not correspond to real clubs were rewritten as readable pseudo-codes (RETIREMENT, WITHOUT_CLUB, UNKNOWN_CLUB, CAREER_BREAK) so that these records can be tracked through the pipeline rather than discarded.

6.3 Conservative competition imputation

Many transfers had a missing competition ID although both club IDs were known. For each club, all competition IDs observed in other transfers are inspected. If exactly one competition ID is observed, it is used for imputation. If multiple competition IDs are observed, the value remains missing. Records whose competition cannot be recovered are labeled UNKNOWN_COMPETITION.

6.4 Strict and with-unknown variants

Two parallel variants of the cleaned data are produced:

- `with_unknown`: retains transfers whose source or destination competition is UNKNOWN_COMPETITION or a pseudo-destination, labelled explicitly.
- `strict`: drops all transfers where either endpoint is unidentified (UNKNOWN_COMPETITION or pseudo-code).

All analyses in this report use the `strict` variant.

The unknown and pseudo endpoints cannot be used meaningfully as nodes in the league-level network and keeping them in the network would inflate the degree and strength of artificial nodes. The volume of identified league-to-league transfers (586,520, see Table 2) is sufficient to support all downstream analyses.

6.5 Edge-list aggregation

For each variant, transfers are aggregated to a directed league-to-league edge list. Two granularities are produced:

- `edge_all_*`: aggregated over all 32 seasons; one row per (source league, destination league) pair.
- `edge_season_*`: per-season; one row per (source league, destination league, season) triple.

Each edge row carries the source and destination league IDs and names, country IDs, tier classifications, the number of transfers (`n_transfers`), the number of loans (`n_loans`), fee sum and median, and the domestic share (the fraction of transfers on that edge with identical source and destination country).

6.6 Competition metadata

A supplementary mapping file (`competition_mapping.csv`) maps each competition ID to a human-readable league name, country, tier (1–4, plus a Youth category), canonical ID for renamed leagues, and prior league names. The file is produced by a separate scraping script,

while mappings for leagues that changed names over time were verified manually using the Transfermarkt website. This is required for tier-based analyses (H2, H4) and for the readable league labels in all figures.

7 Modelling

All modelling is implemented on the strict edge-list variant.

7.1 Graph construction

The aggregated directed weighted graph G_{all} is built from the strict file that contains all edges and aggregates over all seasons. It uses `igraph`, with one node per unique league ID and one edge per (source $\rightarrow$ destination) league pair. The edge weight is the number of player transfers over the 32-season window. A second edge attribute `distance = 1/weight` was also added so that the betweenness-centrality computations interpret heavy flows as short paths. Human-readable league names were used as node labels. The graph G_s was constructed per season in an identical way to analyse the temporal hypotheses H1 and H3. Self-loops, that is, transfers that occurred within one league, were kept in the graph but excluded when computing cross-league metrics, since by construction they are not cross-league transfers. For the null-model rewiring in H3 these loops were removed because they cannot be preserved under degree-preserving edge rewiring.

7.2 Node-level network metrics

For each league in G_{all} the following metrics are computed: weighted in- and out-strength (sum of incoming/outgoing edge weights), in- and out-degree (number of distinct partner leagues), and directed weighted betweenness centrality on the `distance` attribute. To rank the betweenness, shortest paths on inverse weights were used, so that heavy corridors contribute short edges.

7.3 Community detection

To detect communities within the graph the Leiden algorithm [Traag et al., 2019] was used with the

`RBConfigurationVertexPartition` quality function and weighted by the number of transfers. The RB-configuration partition can be used for weighted directed graphs and allows resolution tuning. It is also used for descriptive community structure such as community labels or a meta-graph.

7.4 Null-model

To test the significance of H3, several adjustments were necessary to make the observed and null modularity comparable.

The quality function is switched to `ModularityVertexPartition` which gives the standard modularity Q on a scale that is directly comparable across graphs of similar size. Also

self-loops are removed since the degree-preserving rewiring is not defined on loops. The observed modularity Q^{obs} is then computed on the self-loop-free graph using Leiden with that quality function.

The null distribution is generated by 1,000 degree-preserving random rewirings of the graph. Each rewiring shuffles the edge endpoints of each node while preserving in- and out-degree, using a rewire factor of two times the edge count. Leiden is then applied to each rewired graph, producing Q_i^{null} for each run. The one-sided empirical p -value is computed with an add-one (Laplace) estimator, $p = (\#\{i : Q_i^{\text{null}} \geq Q^{\text{obs}}\} + 1) / (1,000 + 1)$, so that an observed p of zero is reported as $p \approx 0.001$ rather than exactly zero.

The modularity per season is computed in the same way on each G_s to check the temporal stability of these communities. This temporal analysis is limited to the seasons from 2000/01 onwards, where the coverage is dense enough.

7.5 Hypothesis-specific computations

H1 – Concentration. The HHI of inflows is computed as $\text{HHI} = \sum_v (s_v^{\text{in}}/S)^2$ over all destination leagues v , where s_v^{in} is the weighted in-strength and S the total inflow.

The Top-5 share is the sum of the five largest s_v^{in} divided by S . Both are computed on the aggregated graph and per season. The Mann–Kendall test is applied to the two per-season series to test for a monotonic trend over time.

H2 – Bridge leagues. H2 was evaluated in two stages because a CRISP-DM iteration was necessary. In the first stage, the proposal was followed literally: leagues are ranked by directed weighted betweenness centrality on distance. Mid-tier is defined as tier 2–4. Among the top-25 leagues by betweenness, the mid-tier share is compared to the mid-tier base rate across all leagues in the graph. This ratio is the *lift*.

Inspection of this approach revealed two problems. First, betweenness on the count-weighted graph measures topological shortest-path intermediation rather than the stepping-stone role targeted by RQ2. Second, the proposal’s notion of mid-level leagues actually refers to mid-standing leagues in the global hierarchy, not to national second or third divisions: the textbook stepping stones (Portugal, the Netherlands, Belgium) are first divisions in their own countries. The second approach therefore uses a global-standing axis derived from player market values and a flow-based bridge score that explicitly captures upward, cross-regional pass-through.

To estimate a league’s standing, Transfermarkt’s player market-value estimates are used. A league’s standing is the median market value of its incoming transferred players, subject to a minimum of 30 valued incoming transfers (431 leagues qualify). The market values are only used as a node attribute. The edges themselves remain count-weighted, which keeps the project’s count-only weighting consistent with the fee exclusion in Section 5.

The bridge score combines two cross-community flow shares, both computed on the self-loop-free edge list:

- `up_cross_out`: share of a league’s outgoing transfers that go to a higher-standing league in a *different* Leiden community.

- `below_cross_in`: share of a league’s incoming transfers that come from a lower-standing league in a *different* community.
- $\text{bridge_score} = \sqrt{\text{up_cross_out} \times \text{below_cross_in}}$

Using the geometric mean ensures that real stepping stones score high on both factors. Pure hubs or sinks collapse one factor to nearly zero and domestic pyramid churn is excluded because it stays within one community. The test universe is restricted to first-division leagues with throughput $\geq 1,000$ cross-league transfers and known standing (51 leagues), so the contrast is not driven by tiny leagues. The bridge hypothesis predicts an inverted-U of the bridge score against standing, since the apex (Big-5) absorbs talent without passing it on, the bottom produces locally, and the leagues in between route players upward. The fit is a throughput-weighted quadratic in the standing percentile, and significance is tested with a 500-permutation shuffle of the standing labels across the universe leagues (seed = 12412672), producing an empirical null distribution of the quadratic coefficient c_2 .

H4 – Lower-league pathways. Source leagues are grouped into four tiers: Top (tier 1), Second (tier 2), Mid (tier 3–4), Lower (tier 5+). For each group the weighted domestic share is computed as a transfer-count-weighted mean of the per-edge `domestic_share`.

8 Evaluation

This section reports the results obtained from the aggregated and per-season graphs and evaluates each hypothesis against the KPI thresholds defined in Table 1. All values are computed on the strict edge-list variant.

8.1 H1 – Concentration

On the aggregated graph the HHI of inflows is $\text{HHI} = 0.0096$, far below the 0.15 threshold for “concentrated”. The top-5 leagues receive 12.4% of all cross-league inflows, and the top-10 receive 21.2%, both well below the 50% threshold. The five largest inflow destinations are Serie A (Italy), Serie B (Italy), Campeonato Brasileiro Série A (Brazil), the Championship (England) and League One (England) see Figure 1.

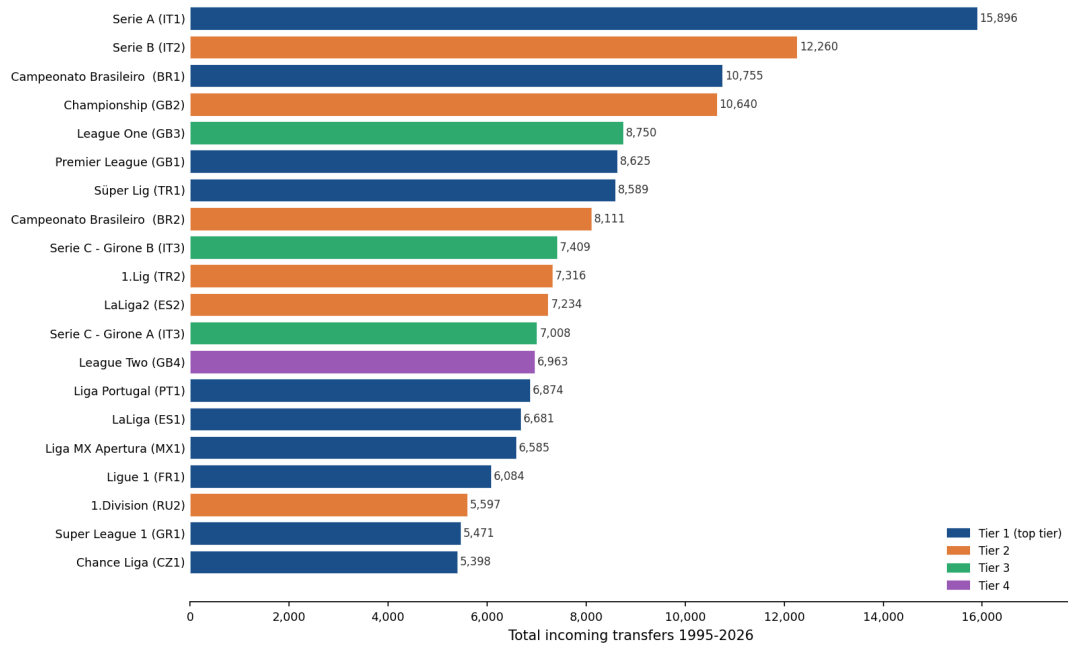


Figure 1: Top 20 leagues by incoming transfers.

The Mann–Kendall test strongly rejects the null hypothesis of no monotonic trend for both metrics, with the HHI showing a decreasing trend with $\tau = -0.93$, $p < 10^{-4}$ and the Top-5 share decreasing with $\tau = -0.92$, $p < 10^{-4}$; see Figure 2. This means that the Top-5 share of incoming transfers declines from roughly 27% in 2000/01 to roughly 9% in 2024/25.

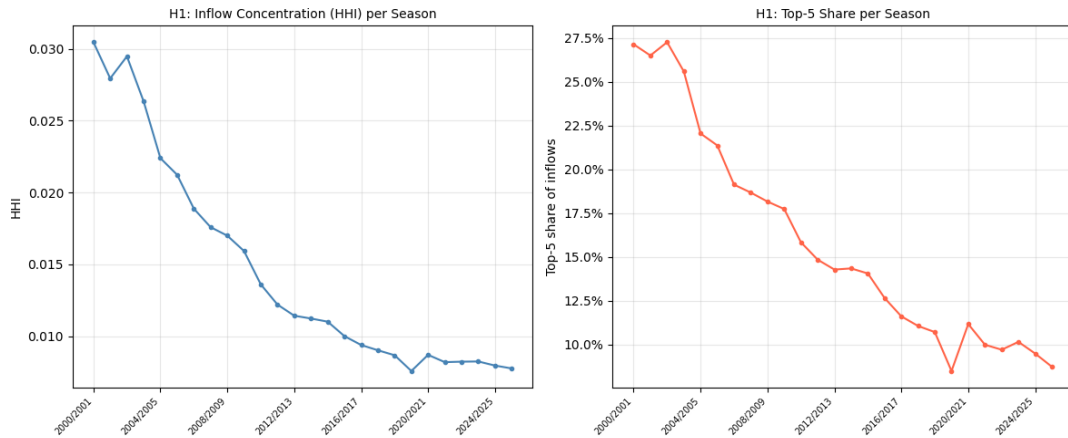


Figure 2: Per-season trends for concentration, 2000/01–2025/26.

Verdict. *H1 is not supported.* The market is moderately spread but does not meet the predefined concentration thresholds. However, concentration declines monotonically across the observation window.

8.2 H2 – Bridge leagues

Initial result. Directed weighted betweenness centrality on the aggregated graph produces a ranking dominated by large top-tier leagues. Among the top-25 leagues by betweenness, the mid-tier (tier 2–4) share is 40.0%, compared to a mid-tier base rate of 40.3% across all 672 leagues (see Figure 3). The resulting lift is $0.99\times$, essentially identical to chance and well below the $1.5\times$ threshold for “supported”.

CRISP-DM iteration. Inspection of this result during the evaluation phase exposed a methodological limitation. Betweenness on a count-weighted graph measures topological shortest-path intermediation rather than the brokerage role the proposal had in mind. On top of that, the original tier-based definition turned out to test a construct nobody actually meant: the proposal’s “mid-level leagues” are first divisions of intermediate global standing (Portugal, Netherlands, Belgium), not national second divisions. In line with the iterative nature of CRISP-DM, the project returned to the modelling phase (Section 7) and introduced a global-standing axis and a flow-based bridge score before re-evaluating the hypothesis.

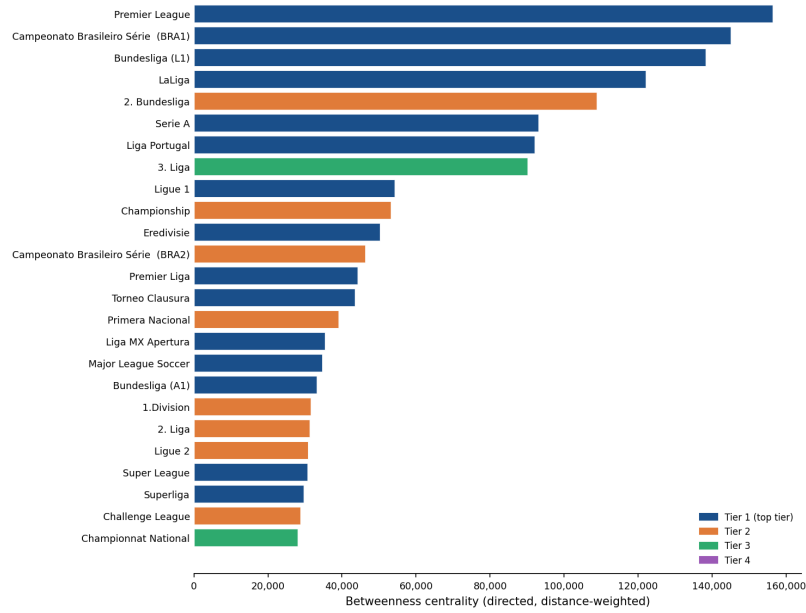


Figure 3: Top 25 leagues by betweenness centrality.

Revised operationalisation. On the 51-league universe of first divisions with throughput $\geq 1,000$ and known standing, the throughput-weighted quadratic fit of bridge score against standing percentile is inverted-U shaped, with $c_2 = -0.53$ and a peak at 52% standing percentile. The tercile means of the bridge score are: low standing = 0.205, mid standing = 0.250, high standing = 0.173. The strongest empirical bridges include data-driven cases such as Bulgaria, Australia, Poland and Romania, alongside textbook classics like Portugal, the Netherlands and Belgium, all sitting on or near the peak of the curve; the Big-5 sit on the declining arm at the right edge, consistent with their role as net apex sinks rather than brokers (Figure 4).

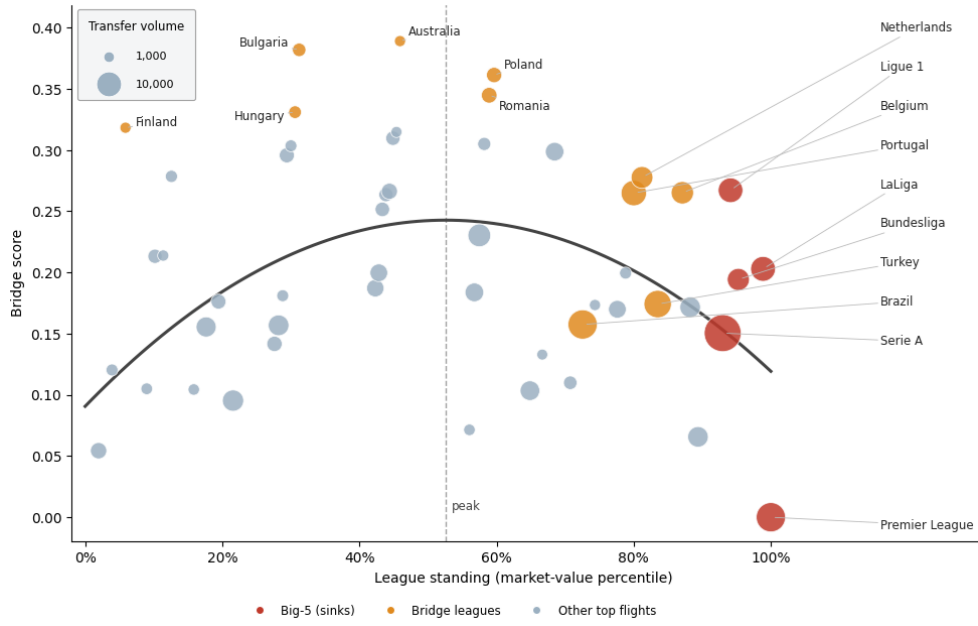


Figure 4: Bridge score against global standing (51 first-division leagues with throughput $\geq 1,000$). The throughput-weighted quadratic fit peaks at the 52% standing percentile.

The 500-run label-permutation test gives an empirical p -value of $p = 0.004$ for the curvature ($c_2 < c_2^{\text{null}}$), well below the $\alpha = 0.05$ threshold. The geometric-mean construction of the bridge score is mechanically low at both extremes, so the permutation test is what shows that the observed mid-standing peak exceeds that mechanical baseline. The raw tercile contrast alone is not significant.

Verdict. *H2 is supported (qualified).* Mid-standing top-flight leagues disproportionately broker upward, cross-regional flows. The apex (Big-5) acts as a net sink and the bottom of the universe produces locally. The effect is moderate in size since it is only one operationalisation, but the original betweenness ranking is retained in the notebook as the first iteration and as an explicit demonstration that operationalisation choice can fully flip the verdict.

8.3 H3 – Community structure

Leiden community detection reveals 17 communities on the aggregated graph when communities with fewer than five leagues are excluded. The communities are geographically coherent. The labels designed by hand include *Italian* (blue), *British Isles* (light blue), *German–Austrian* (green), *Francophone* (brown), *Iberian* (pink), *Pan-American* (orange), *Lusophone & East Asian* (dark green), *Post-Soviet (CIS)* (dark brown), *Balkan & Eastern Mediterranean* (red), *Turkish* (purple), *Nordic–Dutch* (grey), *MENA & African* (yellow), *Swiss* (cyan), *East Asian & Pacific* (light blue), and *Czecho-Slovak* (grey). See Figure 5 for the global map and Figure 6 for the community meta-graph.

The null-model significance test compares the observed modularity against 1,000 degree-preserving random rewirings of the self-loop-free graph (Section 7):

- Observed modularity $Q^{\text{obs}} = 0.2768$

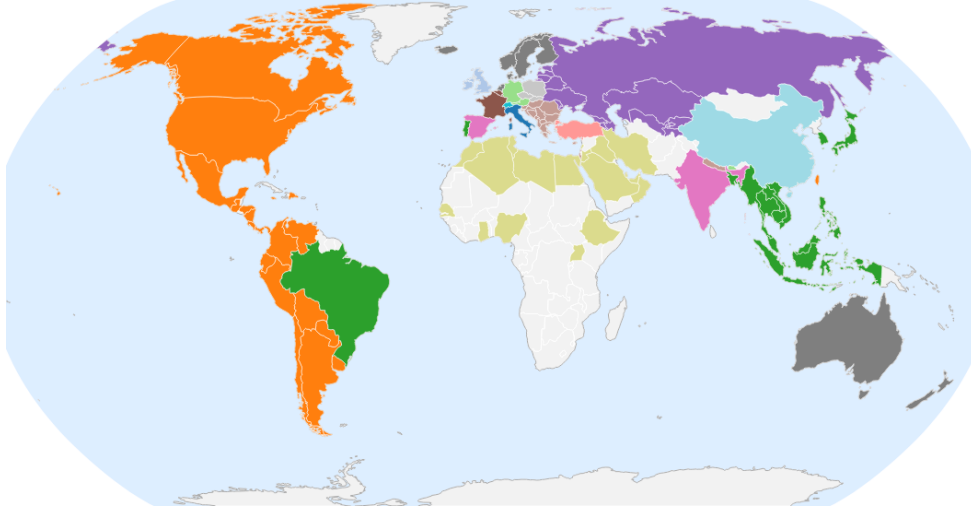


Figure 5: Global Football Transfer communities.

- Null mean modularity $\bar{Q}^{\text{null}} = 0.0678$
- Null standard deviation $\sigma^{\text{null}} = 0.0013$
- Standardised deviation $z = 158.6$
- Empirical one-sided $p \approx 0.001$ with add-one estimator (no null run exceeded Q^{obs})

The per-season analysis (Figure 7) shows that in every one of the 26 seasons examined from 2000/01 onwards, observed Q exceeds the null mean.

Verdict. H3 is *supported*. The transfer network has strong, non-random community structure that is stable across seasons and geographically interpretable.

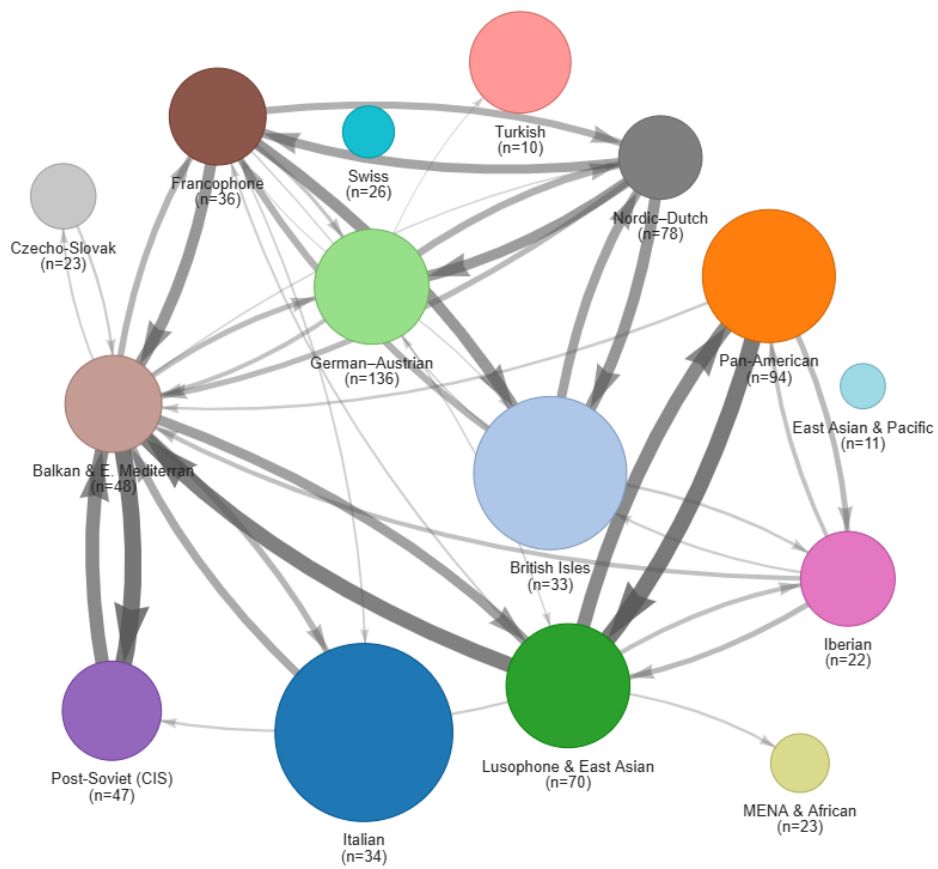


Figure 6: Community meta-graph (non-connected nodes fall under the threshold).

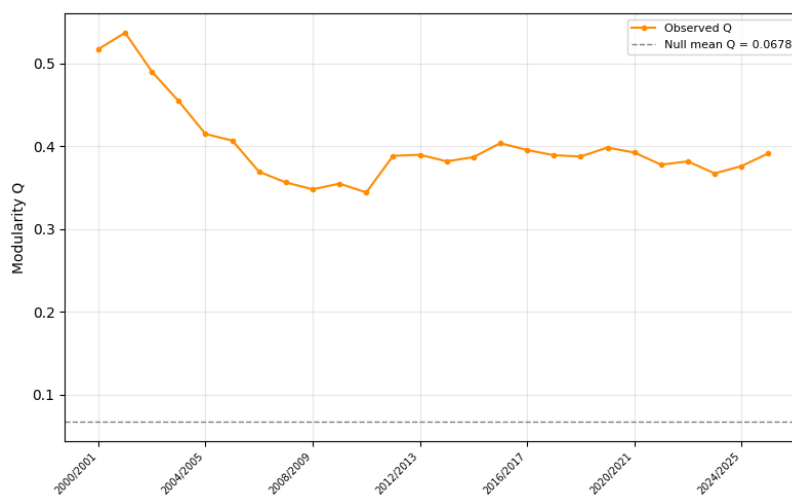


Figure 7: Observed community modularity Q against the null model by season.

8.4 H4 – Lower-league pathways

The transfer-count-weighted domestic share of outgoing transfers increases with source-league tier:

- Top (Tier 1): 56.3% domestic
- Second (Tier 2): 77.8% domestic
- Mid (Tier 3–4): 91.7% domestic
- Lower (Tier 5+): 91.9% domestic

Both mid- and lower-tier shares exceed the 70% threshold by a wide margin (Figure 8). 56.5% of transfers originating in Tier 5+ source leagues are upward (to a higher tier).

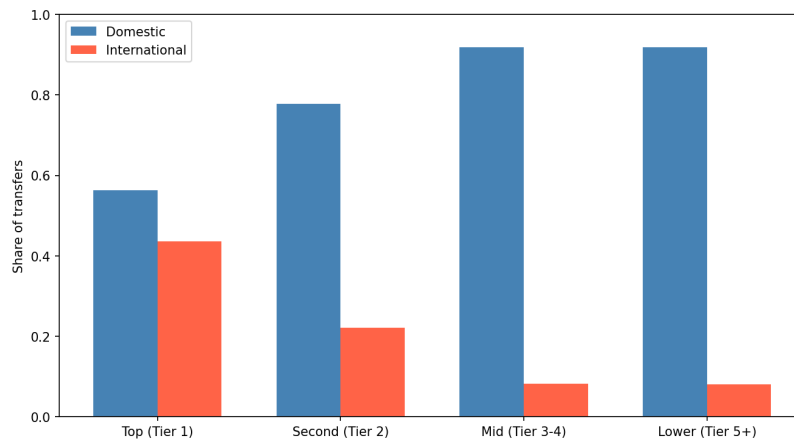


Figure 8: Domestic vs. international transfer share by source tier.

Verdict. *H4 is supported.* Lower-tier leagues function predominantly as national labour markets, unlike higher-tier leagues.

8.5 Summary against KPIs

H1 – Concentration.

NOT SUPPORTED (DECLINING TREND).

Threshold: $HHI > 0.15$ or $Top-5 > 50\%$.

Result: $HHI = 0.0096$, $Top-5 = 12.4\%$. Mann–Kendall: strong monotonic decline ($\tau = -0.93$, $p < 10^{-4}$).

H2 – Bridge leagues.

SUPPORTED (QUALIFIED, AFTER ITERATION).

Threshold: curvature $c_2 < 0$ and $p < 0.05$ on the revised bridge-score fit.

Result: Inverted-U fit $c_2 = -0.53$, peak at 52% standing percentile; tercile means low/mid/high = 0.205/0.250/0.173; permutation $p = 0.004$. Original betweenness ranking gave lift = $0.99\times$, not supported.

H3 – Community structure.

SUPPORTED.

Threshold: $p < 0.05$.

Result: $Q^{obs} = 0.277$ vs. $\bar{Q}^{null} = 0.069$; $z = 158.6$, $p \approx 0.001$.

H4 – Lower-league pathways.

SUPPORTED.

Threshold: domestic share $> 70\%$ for tier 3+.

Result: Tier 3–4 = 91.7% domestic, Tier 5+ = 91.9% domestic.

9 Deployment

The final result of this project is a reproducible pipeline. This section therefore presents the produced results, where to find them in the project and how to reproduce them end-to-end. The complete project is available at <https://github.com/timg4/Mobility-in-Global-Football>.

The pipeline itself is split into two notebooks `01_data_preparation.ipynb` and `02_analysis.ipynb` that are to be executed in sequence. When done so, they produce the cleaned edge-list CSVs, the result figures and two final summary tables.

Reproducibility is supported by having pinned Python dependencies; a fixed random seed with seed = 12412672 for all random experiments such as the Leiden partitioning; and both notebooks use relative paths to the project root, so rerunning them would produce the exact same figures and numbers.

Three figures are additionally produced as interactive HTML files. The community meta-graph can be explored by hovering over the communities, which reveals their league members and the size of each cross-community flow. The community map is also represented in an HTML choropleth of the dominant communities per country. The static PNGs embedded in the report show the overall shape, while these interactive visualisations allow users to drill down into individual leagues and corridors.

10 Conclusion and Outlook

This project created a directed, weighted league-to-league network over 32 seasons, built from 586,520 league-to-league transfer records across 672 leagues in 118 countries and used it to test four hypotheses.

Three of these hypotheses could be supported by the data and one was not. The network is not concentrated in the sense of the proposal’s thresholds ($\text{HHI} = 0.0096$, Top-5 share = 12.4%). The original H2 operationalisation by betweenness centrality and national tier did not support the bridging hypothesis (lift = $0.99\times$ vs. baseline), but a revised analysis on a global-standing axis revealed an inverted-U pattern (curvature $c_2 = -0.53$, permutation $p = 0.004$). Mid-standing top-flight leagues, both textbook classics such as Portugal, the Netherlands and Belgium and data-driven cases such as Bulgaria, Australia, Poland and Romania, do broker upward, cross-regional player flows. The market also shows strong, geographically coherent community structure ($Q = 0.277$, $z = 158.6$, $p \approx 0.001$ against a degree-preserving null model with 1,000 rewirings) that is stable across seasons. The detected communities map clearly onto linguistic, regional and especially historical ties. The data also suggests that lower-tier leagues behave overwhelmingly as national labour markets, with domestic transfer shares above 91% for tier 3 and below. Another notable finding is that inflow concentration has declined significantly over the entire observation window ($\tau = -0.93$, $p < 10^{-4}$), which reflects the globalisation of the player labour market. Loans were treated

as ordinary edges in the main analysis; their distinct behaviour as temporary, downward and largely returning moves is explored separately in Appendix A.

Limitations

This work has several limitations. Transfer fees were excluded from the analysis because of their non-random missingness. An analysis restricted to leagues with reliable fee coverage would complement the count-based view used here, particularly for the economically strong top-tier European leagues. In addition, loans were counted as ordinary transfers in the H1–H4 analyses although they are structurally different from permanent transfers; an exploratory loan analysis is provided in Appendix A, but a full integration of the loan/permanent distinction into the core hypotheses remains future work. Furthermore, the strict variant removes all transfers into pseudo-destinations such as retirement or clublessness. These exits are arguably also a form of mobility, and dropping them mechanically inflates the domestic and upward shares in H4: a lower-tier player who retires disappears from the statistics instead of counting as an outgoing move. Finally, the global-standing axis used in the revised H2 analysis depends on Transfermarkt’s own market-value estimates, which have somewhat lower coverage for small or non-European leagues. The median is robust to this and these leagues correctly land at the bottom of the standing scale, but a fully-fledged fee- or value-weighted edge analysis remains future work.

Outlook

Future work could combine the league-level network from this project with club-level networks of prior work [Liu et al., 2016, Li et al., 2019, Matesanz et al., 2018, Wand, 2022, D’Urso et al., 2025]. This would result in a two-layer model in which clubs are nested within their league communities and league-level flows constrain club-level mobility. A second direction would be to incorporate player attributes such as nationality, age or position to surface structural findings on *which* players are moving. A third follow-up would be a fee-aware analysis restricted to leagues with reliable fee coverage, as mentioned in the Limitations.

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A Loans

Loans were treated as ordinary edges in H1–H4, but a loan is conceptually different from a permanent transfer: it is temporary, the player is expected back at the parent club, and it is often used for player development. The Transfermarkt data flags loans directly (`ACTIVE_LOAN_TRANSFER` for the outgoing move and `RETURNED_FROM_PREVIOUS_LOAN` for the return), which makes it possible to follow each loaned player and ask three questions: do they return to the parent club, does the loan cross a border, and which direction along the league pyramid does it go?

Method. For each transfer flagged as `ACTIVE_LOAN_TRANSFER` in the analysis window, the player’s subsequent transfer history is inspected for a `RETURNED_FROM_PREVIOUS_LOAN` back to the source club and for a later `STANDARD` move to the loan club (a loan-to-buy outcome). Each loan episode is also classified by direction along the tier hierarchy of the source and destination league (downward, same level, or upward) and by whether the move crosses a national border. The implementation reads the raw transfer table directly from the SQLite database, since the aggregated edge lists do not preserve the individual player trajectories needed for episode tracking.

Findings. The analysis identifies 126,364 loan episodes within the analysis window. Note that the 126,364 loan episodes differ from the 249,985 loan-related transfer records in Table 2, since the latter count both the outgoing loan and its return as separate records. Loans behave as a sharply different mode of mobility from permanent transfers:

- **Return rate.** 95% of loan episodes are followed by a return to the parent club, with very little difference between domestic loans (95%) and international loans (96%). The median loan duration is 181 days, which lines up with the standard half-season loan window in most leagues.
- **Direction along the pyramid.** 50% of loans go downward (to a lower division), 41% stay at the same level, and only 9% go upward. This is the mirror image of the permanent lower-tier pattern from H4, where transfers originating from lower tiers are predominantly upward.
- **Domestic share.** 79% of loans are domestic, sitting between the H4 domestic shares for Tier-1 (56%) and Tier-2 (78%) permanent transfers.
- **Loan-to-buy.** 15% of loan episodes are later followed by a permanent move from the parent club to the same loan club, suggesting that a meaningful minority of loans function as trial periods rather than pure development moves.

Interpretation. Loans complement the H4 finding rather than contradict it. Permanent lower-tier moves are domestic and upward because they reflect actual career progression. Loans, by contrast, are predominantly domestic and *downward*: a player owned by a higher-tier club is sent to a lower-tier club to gain playing time, then returns. Treating both as ordinary edges in H1–H4 mixes two structurally different mobility modes, which is a known limitation of the main analysis (Section 10). A fuller integration of the loan/permanent distinction into the core hypotheses – in particular a re-evaluation of H4 directional shares on permanent-only edges – is identified as future work.

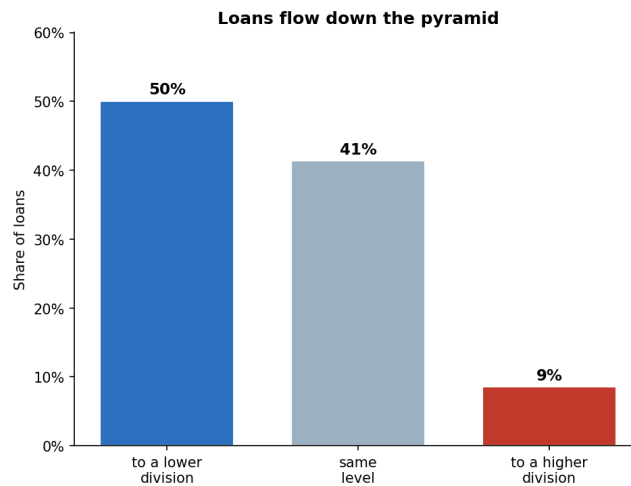


Figure 9: Direction of loan episodes along the league tier hierarchy. Loans flow predominantly downward.

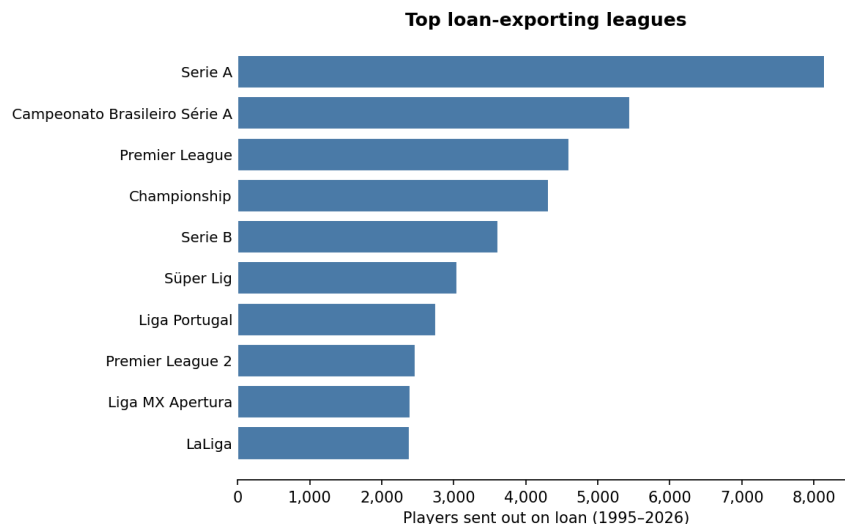


Figure 10: Top 10 loan-exporting leagues by absolute number of outgoing loan episodes, 1995–2026.